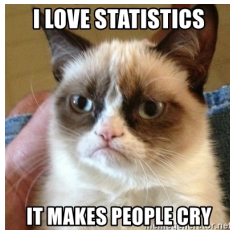


Cox Models for Political Science:

An Introduction to Duration (Survival) Analysis

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Outline

1. Introduction to Duration Analysis
2. Cox Proportional Hazard Model
3. Example 1: Duration in Authoritarian Regime
4. Example 2: Veterans' Administration Lung Cancer Dataset

Introduction to Duration Analysis

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What is survival in Political Science?

- We want to study duration of political events or institutions.
- For example:
 - How long a regime survives before collapsing?
 - How long a cabinet remains in office?
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	cowcode	year	gwf_country	gwf_casename	gwf_startdate	gwf_enddate	gwf_spell	gwf_duration	gwf_fail
	<int>	<int>	<chr>	<chr>	<chr>	<chr>	<int>	<int>	<int>
1	40	1953	Cuba	Cuba 52-59	1952-03-10	1959-01-01	7	1	0
2	40	1954	Cuba	Cuba 52-59	1952-03-10	1959-01-01	7	2	0
3	40	1955	Cuba	Cuba 52-59	1952-03-10	1959-01-01	7	3	0
4	40	1956	Cuba	Cuba 52-59	1952-03-10	1959-01-01	7	4	0
5	40	1957	Cuba	Cuba 52-59	1952-03-10	1959-01-01	7	5	0
6	40	1958	Cuba	Cuba 52-59	1952-03-10	1959-01-01	7	6	0
7	40	1959	Cuba	Cuba 52-59	1952-03-10	1959-01-01	7	7	1

Cox Proportional Hazard Model

Key Features of the Cox Model (1/2)

- The Cox model answers the question:
 - **Which variables (such as education, age, gender, etc.) affect the hazard rate?**

In other words...

The Cox model estimates how certain characteristics accelerate or decelerate the timing of an event's occurrence.

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- The hazard rate for the i -th individual is defined as

$$h_i(t) = h_0(t) \exp(\beta' X_i) \quad (1)$$

where $h_0(t)$ is the **baseline hazard**,
and $\beta' X_i$ represents the **covariates** and **regression parameters**.

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! Important

In Cox model, the main *Quantity of Interest* we care about is *the hazard rate* ($h_i(t)$).

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 - It does not include an intercept term.
- In *parametric* models, we apply **Maximum Likelihood Estimation (MLE)** to maximize the likelihood function and obtain the parameter estimates that best fit the data.
- In *semi-parametric* models, such as Cox model, we rely on *Partial* Likelihood Estimation, as we can only maximize part of the parameters, while leaving $h_0(t)$ is unspecified.

Example 1: Duration in Authoritarian Regime

Fit Cox Model

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- **Treatment:**

- `gwf_military`, coded as 1 if a country is classified as a military regime, otherwise 0.

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- **Treatment:**

- `gwf_military`, coded as 1 if a country is classified as a military regime, otherwise 0.

```
# library {survival}
library(survival)

# Fit cox model
model <- coxph(
  Surv(gwf_duration, gwf_fail) ~ gwf_military + wdi_gdpcapcur + wdi_pop,
  data = new_df
)
```

How to Read the Cox Model Output

```
summary(model)
```

```
Call:
```

```
coxph(formula = Surv(gwf_duration, gwf_fail) ~ gwf_military +  
      wdi_gdpcapcur + wdi_pop, data = new_df)
```

```
n= 3141, number of events= 157
```

```
(1450 observations deleted due to missingness)
```

	coef	exp(coef)	se(coef)	z	Pr(> z)	
gwf_military	2.231e+00	9.307e+00	1.916e-01	11.642	< 2e-16	***
wdi_gdpcapcur	-5.076e-04	9.995e-01	9.224e-05	-5.503	3.73e-08	***
wdi_pop	-7.013e-09	1.000e+00	3.263e-09	-2.149	0.0316	*

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

	exp(coef)	exp(-coef)	lower .95	upper .95
gwf_military	9.3072	0.1074	6.3932	13.5494
wdi_gdpcapcur	0.9995	1.0005	0.9993	0.9997
wdi_pop	1.0000	1.0000	1.0000	1.0000

Counterfactual Prediction (1/6)

```
library(marginaleffects)

min_time <- min(new_df$gwf_duration[new_df$gwf_fail == 1], na.rm = TRUE)
max_time <- max(new_df$gwf_duration[new_df$gwf_fail == 1], na.rm = TRUE)

nd <- datagrid(
  gwf_military = 0:1,
  gwf_duration = round(seq(min_time, max_time, length.out = 25)),
  grid_type = "counterfactual",
  model = model
)
```

- ① Specify groups to compare (military vs. non-military regimes)
- ② 25 equally spaced points in time between the first (`min_time`) and the last (`max_time`) occurring event time
- ③ Tells the function to hypothesize specific values of predictors (`gwf_military == 1` or `0`), regardless of the observed values in the original dataset.
- ④ Specify the model used to generate predictions.

Counterfactual Prediction (2/6)

```
p <- avg_predictions(model,  
                      type = "survival",  
                      by = c("gwf_duration", "gwf_military"),  
                      vcov = "rsample",  
                      newdata = nd)
```

- ① Choose the survival probability scale.
- ② Calculate probability for each combination of `gwf_military` and `gwf_duration`. Every probability represents a point (survival probability).
- ③ Non-parametric bootstrapping.
- ④ Feed `avg_prediction()` with the new data that we just simulated.

Counterfactual Prediction (3/6)

```
tail(p)
```

	gwf_duration	gwf_military	estimate	conf.low	conf.high
45	96	0	0.7365691	0.5827513	0.8514241
46	96	1	0.1708886	0.1009036	0.3331206
47	101	0	0.7365691	0.5827513	0.8514241
48	101	1	0.1708886	0.1009036	0.3331206
49	105	0	0.7365691	0.5827513	0.8514241
50	105	1	0.1708886	0.1009036	0.3331206

Counterfactual Prediction (4/6)

- It is useful to factor variables before plotting

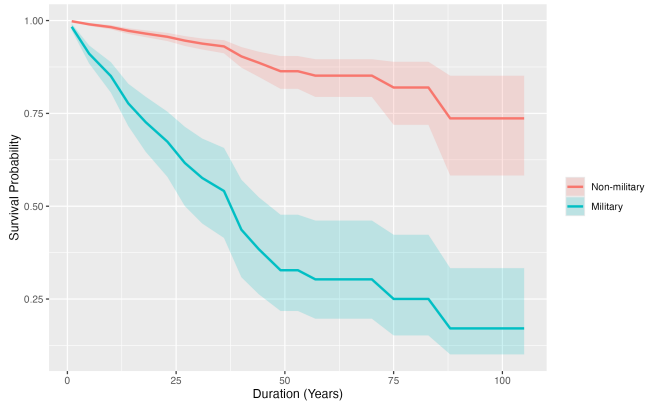
```
# Factoring treatment variable
# library(dplyr)
p <- p |>
  mutate(gwf_military = factor(gwf_military,
                                levels = c(0, 1),
                                labels = c("Non-military", "Military")))
```

Counterfactual Prediction (5/6)

```
library(ggplot2)

# Apply ggplot()
ggplot(p, aes(x = gwf_duration, y = estimate,
              color = as.factor(gwf_military),
              fill = as.factor(gwf_military))) +
  geom_line(linewidth = 1) + # Step 1: Draw survival lines
  geom_ribbon(aes(ymin = conf.low, ymax = conf.high),
             alpha = 0.2, color = NA) + # Step 2: Add confidence bands
  labs(
    x = "Duration (Years)",
    y = "Survival Probability",
    color = "",
    fill = ""
  ) # Step 3: Clean axis labels and legend
```


Counterfactual Prediction (6/6)



Differences in Predicted Survival Probability (1/3)

- Difference means that:
 - $\mathbb{E}(\textit{Survival} \mid \textit{Military}) - \mathbb{E}(\textit{Survival} \mid \textit{Non_Military})$

Differences in Predicted Survival Probability (1/3)

- Difference means that:

- $E(\text{Survival} \mid \text{Military}) - E(\text{Survival} \mid \text{Non_Military})$

```
tmax <- quantile(new_df$gwf_duration, 0.95, na.rm = TRUE) ①

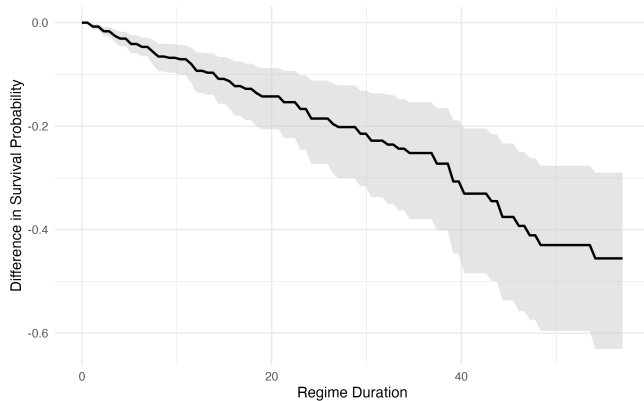
# Generate Predictions
m <- comparisons(
  model,
  variables = "gwf_military", ②
  newdata = datagrid(gwf_duration = seq(0, tmax, length.out = 100)), ③
  type = "survival", ④
  vcov = "rsample" ⑤
)
```

- ① Use the 95th percentile of regime duration to avoid extreme outliers.
- ② Compare military vs. non-military regimes.
- ③ Create a grid of duration values from 0 to tmax
- ④ Predict survival probabilities over time.
- ⑤ Use bootstrap (resampling) to estimate more robust confidence intervals.

Difference in Predicted Survival Probability (2/3)

```
ggplot(m, aes(x = gwfd_duration, y = estimate)) +  
  geom_ribbon(aes(ymin = conf.low, ymax = conf.high),  
             fill = "grey80", alpha = 0.5) +  
  geom_line(size = 1, color = "black") +  
  labs(x = "Regime Duration",  
       y = "Difference in Survival Probability") +  
  theme_minimal(base_size = 13)
```

Difference in Predicted Survival Probability (3/3)



Example 2: Veterans' Administration Lung Cancer Dataset

Fit Cox Model with `coxph()`

- Fit Cox Model with Veterans
 - Time variable: `Time`
 - Event indicator variable: `Status`
 - Treatment: `trt`

Fit Cox Model with `coxph()`

- Fit Cox Model with Veterans
 - Time variable: Time
 - Event indicator variable: Status
 - Treatment: trt

```
# Load and prepare the data  
veteran <- survival::veteran
```

```
# Fit Cox proportional hazards model  
model_c <- coxph(Surv(time, status) ~ trt, data = veteran)  
summary(model_c)
```


Counterfactual Predictions with `marginalEffects()` (1/3)

- Use `avg_predictions()` to estimate average predicted survival at each time point for each treatment group.
- Also use `vcov = "rsample"` to obtain bootstrapped standard errors.

Counterfactual Predictions with `marginalEffects()` (2/3)

```
library(marginalEffects)

min_time <- min(veteran$time[veteran$trt == 2], na.rm = TRUE)
max_time <- max(veteran$time[veteran$trt == 2], na.rm = TRUE)

nd <- datagrid(
  status = 0:1,
  time = round(seq(min_time, max_time, length.out = 200)),
  grid_type = "counterfactual",
  model = model_c
)
```

- ① Specify groups to compare (chemotherapy vs. standard)
- ② 100 equally spaced points in time between the first (`min_time`) and the last (`max_time`) occurring event time
- ③ Tells the function to hypothesize specific values of predictors (`trt == 1` or `2`), regardless of the observed values in the original dataset.
- ④ Specify the model used to generate predictions.

Counterfactual Predictions with `marginalEffects()` (3/3)

```
ap_surv <- avg_predictions(model_c,  
                           type = "survival",  
                           by = c("time", "trt"),  
                           vcov = "rsample",  
                           newdata = nd)
```

```
tail(ap_surv)
```

	time	trt	estimate	conf.low	conf.high
195	979	1	0.023418828	0.004668523	0.04676991
196	979	2	0.021896712	0.001999662	0.05977180
197	989	1	0.023418828	0.004668523	0.04676991
198	989	2	0.021896712	0.001999662	0.05977180
199	999	1	0.005365124	0.001153153	0.01426689
200	999	2	0.004885817	0.001413887	0.00758126

Plot Counterfactual Predictions (1/2)

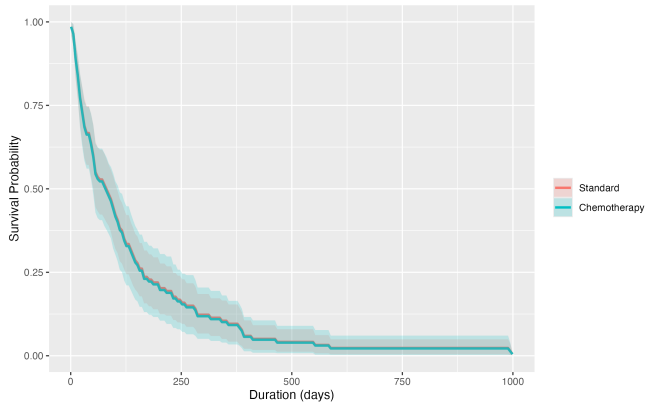
- Similarly, let's factor treatment variable before plotting.

```
# Factoring treatment variable
# library(dplyr)
ap_surv <- ap_surv |>
  mutate(trt = factor(trt,
                      levels = c(1, 2),
                      labels = c("Standard", "Chemotherapy")))
```

Plot Counterfactual Predictions (2/2)

```
ggplot(ap_surv, aes(x = time, y = estimate,
                    color = as.factor(trt),
                    fill  = as.factor(trt))) +
  geom_line(linewidth = 1) + # Step 1: Draw survival lines
  geom_ribbon(aes(ymin = conf.low, ymax = conf.high),
             alpha = 0.2, color = NA) + # Step 2: Add confidence bands
  labs(
    x = "Duration (days)",
    y = "Survival Probability",
    color = "",
    fill  = ""
  )
```

Plot Counterfactual Predictions



Counterfactual Comparisons with `marginalEffects()`

- We can also compare the average survival probability at different and specific time points. We use `comparison()` here and specify the time points we want in the `datagrid()` argument.
 - We generate a new dataset called `newdata` using `datagrid(time = c(100, 200, 300, 400, 500))`.
 - Also Specify the `type = "survival"`
 - What does this mean?

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 - We generate a new dataset called `newdata` using `datagrid(time = c(100, 200, 300, 400, 500))`.
 - Also Specify the `type = "survival"`
 - What does this mean?

```
# Compare at specific time points
library(marginaleffects)
time_comp <- comparisons(
  model_c,
  variables = "trt",
  newdata = datagrid(time = c(100, 200, 300, 400, 500)),
  type = "survival"
)
```


Counterfactual Comparisons with `marginalEffects()`

```
time_comp
```

time	Estimate	Std. Error	z	Pr(> z)	S	2.5 %	97.5 %
100	-0.00646	0.0658	-0.0982	0.922	0.1	-0.1355	0.1225
200	-0.00580	0.0590	-0.0982	0.922	0.1	-0.1215	0.1099
300	-0.00453	0.0461	-0.0982	0.922	0.1	-0.0949	0.0858
400	-0.00294	0.0299	-0.0982	0.922	0.1	-0.0616	0.0557
500	-0.00230	0.0234	-0.0982	0.922	0.1	-0.0482	0.0436

Term: trt

Type: survival

Comparison: +1